OMIS 105 — Introduction to Database Management Systems

Santa Clara University · Leavey School of Business · Department of Information Systems and Analytics
Fall 2026 · 10 weeks, 20 sessions

This syllabus collects, in one document, what the course repository publishes in its separate pages. If this page and a linked page ever disagree, the linked page in `course_information/` is the one that applies, and the instructor will correct this page.

1. Course Information

Course	OMIS 105 — Introduction to Database Management Systems
Units	*See the SCU course listing*
Quarter	Fall 2026
Format	10 weeks · 20 sessions (2 per week) · 100 minutes per session
Meetings	Tuesday and Thursday — section 1: 12:10–1:50 pm; section 2: 2:00–3:40 pm (Pacific Time)
First / last class	*See Camino*
Classroom	*See Camino*
Prerequisite	OMIS 30 — Introduction to Programming
Learning management system	Camino (Canvas) — announcements, Discussions, exams
Course materials	Course repository: https://github.com/mahmoudparsian/OMIS-105 — or one ZIP per week (`DOWNLOADING_COURSE_FILES.md`)

Instructor

Instructor	Dr. Mahmoud Parsian, Assistant Teaching Professor (`INSTRUCTOR.md`)
Email	mparsian@scu.edu
Office hours	Mondays and Wednesdays, 4:15–5:00 pm Pacific Time
Where	In person at Lucas Hall, Room 316U, or on Zoom (link in `QUESTIONS_and_OFFICE_HOURS.md`)
First / last office hour	Wednesday, September 23, 2026 / Wednesday, December 2, 2026

2. Course Description

> You will learn the fundamentals of modern relational data management. Topics include SQL, schema design, data modeling, querying data with SQL, database applications, and transactions. Through lectures, hands-on labs, and assignments, you will discover how real-world database systems work, the principles behind them, and how they shape the way organizations store, retrieve, and analyze information every day.

The course covers the core issues in databases and database management systems (DBMS). Students acquire technical and managerial skills in planning, analysis, design, implementation, and maintenance of databases, with hands-on training in relational database design, normalization, SQL, and database implementation. Use of a DBMS — **DuckDB** — is required; it runs inside Python with no server to install.

Audience: undergraduate students in the Department of Information Systems and Analytics. No prior experience with SQL or databases is needed.

---

3. Learning Outcomes

By the end of this course, students will be able to:

1. **Design** a normalized relational database from business requirements.
2. **Write** complex SQL queries to extract business insights.
3. **Rank** and compare rows with window functions (`ROW_NUMBER`, `RANK`, `DENSE_RANK`).
4. **Manage** data integrity through transactions and constraints.
5. **Answer** multi-step business questions with joins and subqueries.
6. **Communicate** database designs using ER diagrams.

---

4. Course Schedule

Two sessions per week. Each session is a lecture and live demo, then an in-class lab (posted after about 45 minutes of lecture).

Week	Topic	Key SQL concepts
1	Database Foundations	`SELECT`, `WHERE`, `ORDER BY`, `LIMIT`, `LIKE`, `DISTINCT`
2	Relational Modeling	`GROUP BY`, `HAVING`, multi-table schemas
3	SQL Basics	`CASE`, string functions, `UPPER`, `YEAR`, computed columns
4	SQL Aggregation	`COUNT`, `SUM`, `AVG`, `MIN`, `MAX`, `INNER JOIN`, `LEFT JOIN`
5	SQL Joins	Multi-table `JOIN`, `LEFT JOIN`, `IS NULL`, `COALESCE`
6	Joins II	`INNER` / `LEFT` / `RIGHT JOIN` practice, finding what is missing, self-joins, 3+ tables
7	Subqueries	Scalar subqueries, `IN` / `NOT IN` (and the `NULL` trap), `EXISTS`, subqueries in `FROM`, correlated subqueries, CTEs
8	Database Design	Normalization (1NF, 2NF, 3NF), decomposition
9	Ranking Functions	`ROW_NUMBER`, `RANK`, `DENSE_RANK`, `PARTITION BY`, filtering a rank with a CTE
10	Transactions & ACID	`BEGIN`, `COMMIT`, `ROLLBACK`, `CHECK`, `NOT NULL`
Bonus	*Optional:* Project Integration · Review & Modern Data · Query Performance	See `weekly_lectures/bonus/`

Exam	Date and time	Where
Midterm Exam	Thursday, October 22, 2026, in class	Your classroom

Exam	Date and time	Where
Final Exam — 12:10–1:50 pm section	Tuesday, December 8, 2026, 1:30–4:30 pm	129 Vari Hall
Final Exam — 2:00–3:40 pm section	Thursday, December 10, 2026, 1:30–4:30 pm	129 Vari Hall

Source: `ASSIGNMENTS_and_GRADING.md` §2 and `FINAL_EXAM.md`; the university's Fall 2026 final exam schedule.

The topic table is the course's authoritative list (repository `README.md` §18); the slide decks in `weekly_lectures/` and the 10-week outline follow a somewhat slower pacing, and where they disagree, the table applies. Each week's lectures and review notebooks are in `weekly_lectures/` and `weekly_reviews/`.

---

5. Required Materials

No textbook is required. All course materials — lectures, labs, installation guides, reference documents — are in the course repository. Optional reading: *DuckDB in Action*, in the `books/` folder (`REQUIRED_BOOK.md`).

A laptop at every session (fully charged, with its charger): every session includes an in-class lab. A dead battery, a forgotten charger, or software installed incorrectly is not an excused reason to miss a lab (`LAPTOP.md`).

Software — install before the first lab, following the step-by-step guide for macOS or Windows in `software_installation/`:

Tool	Used for
Python 3	Runs the notebooks
DuckDB	The course's SQL database
Marimo	Reactive Python notebooks — every lab is a Marimo notebook
Pandas, Matplotlib	Query results as data frames, and charts
qStudio	Desktop SQL editor
Respondus LockDown Browser	**Midterm and Final Exam** — install it as soon as possible

---

6. Assessment and Grading

Component	Points
20 in-class labs — one per class session, 30 points each	600
Midterm Exam — in class, closed book, LockDown Browser	200
Final Exam — in class, closed book, LockDown Browser	200
Total	**1,000**

Labs

- One lab per class session (20 labs). Each lab is a Marimo notebook (`.py`) — open it in Marimo, complete the exercises, and run all cells.
- The lab is **started, completed, and submitted during the class session** in which it is assigned.

- **Lowest lab replaced:** the lowest lab score is replaced by the average of the other 19 lab scores, so labs still total 600 points.
- Clear SQL logic and problem-solving approach are graded more heavily than minor formatting or syntax issues.
- Each lab (with blanks) and its answers are posted in `LABS/` after class.

Exams

Both exams are **closed book, closed notes, closed software, and closed Internet**, and are taken in the **Respondus LockDown Browser** — install it as soon as possible. Dates: Section 4.

Letter grades

Total points out of 1,000; fractional totals are rounded to the nearest whole point first (939.5 → 940 → A; 939.4 → 939 → A–).

Letter	Points	Letter	Points	Letter	Points
A	940 – 1,000	B–	800 – 829	D+	670 – 699
A–	900 – 939	C+	770 – 799	D	630 – 669
B+	870 – 899	C	730 – 769	D–	600 – 629
B	830 – 869	C–	700 – 729	F	below 600

Full policy: `ASSIGNMENTS_and_GRADING.md`.

---

7. Lab Submission

Item	Rule
When	Before the end of the class session in which the lab is assigned
File name	`week##_lab_##_firstname_lastname.py`, all lower case — e.g. Jane Smith, Week 1: `week01_lab_01_jane_smith.py` (Tuesday) and `week01_lab_02_jane_smith.py` (Thursday)
Late or incomplete	May receive reduced credit unless otherwise approved by the instructor

Full rules: `LAB-IN-CLASS-POLICY.md`.

---

8. Attendance

Attendance is mandatory. Each in-class lab must be completed and submitted during the class session in which it is assigned. If you are absent, you receive 0 on that lab (30 points); a missed lab cannot be made up, because the lab environment and instructor support are only available during class. The "lowest lab replaced" policy absorbs one missed lab. Details: `CLASS_ATTENDANCE.md`.

---

9. Accessibility and Accommodations

If you have a documented disability for which accommodations may be required in this class, contact the **Office of Accessible Education (OAE)** — https://www.scu.edu/oae/ — as early as possible to arrange accommodations, and share your accommodation letter with the instructor.

10. Academic Integrity and Conduct

- **Your own work (NO-COPY-PASTE).** You may use any appropriate resources to help you understand and complete your labs, but the work you submit must reflect your own understanding and effort. You may discuss concepts during class; copying another student's notebook or submitting duplicated work is not permitted.
- **SCU Academic Integrity:** https://www.scu.edu/academic-integrity/ — the university's pledge, policy, and resources.
- **Classroom conduct:** respect, resourcefulness, honesty, and constructive feedback — see `ACADEMIC_CONDUCT.md`.
- **Discrimination, harassment, and sexual misconduct:** Santa Clara University does not tolerate them. Information, reporting options, and support: https://www.scu.edu/title-ix/.

11. Getting Help

Need	Where
A question about course content	Ask in class, come to office hours, post on **Camino Discussions** (students, the TA, and the instructor respond), or email the instructor
Installing the course software	`software_installation/`
LockDown Browser or other SCU technology	SCU Technology Help: (408) 554-5700, or a Computing Support Request

12. Course Documents

Document	Contents
`ASSIGNMENTS_and_GRADING.md`	Grading, exam dates, course policies, letter grades
`LAB-IN-CLASS-POLICY.md`	How the in-class labs work, and file naming
`CLASS_ATTENDANCE.md`	Attendance
`FINAL_EXAM.md`	Final Exam dates, times, and room
`QUESTIONS_and_OFFICE_HOURS.md`	Office hours and how to ask questions
`LAPTOP.md`	Laptop requirement
`REQUIRED_BOOK.md`	Course materials and the optional book
`ACADEMIC_CONDUCT.md`	Classroom conduct
`DOWNLOADING_COURSE_FILES.md`	One ZIP per week (no git needed)

Document	Contents
`outline-10-weeks/`	The 10-week course outline

OMIS 105 — Introduction to Database Management Systems — Fall 2026